Overlay ring-width curves on A3 transparencies

Student PDF: reference and Sample A (2 pages) · Instructor answer PDF

Preparation

Print page 1 (the reference curve) on A3 landscape paper and page 2 (Sample A) on A3 landscape transparency film. Print both at 100% / actual size and disable fit-to-page scaling. Measure each 100 mm check bar with a ruler and confirm it measures 100 mm. Lay the Sample A film over the paper reference and move it horizontally. A white backing sheet helps visibility. The horizontal scale is 3 mm per year.

Activity

1. Overlay dashed Sample A on the solid reference curve. Align equal values on the vertical scales.

2. Slide the sample horizontally to find a sustained match in narrow- and wide-ring patterns.

3. Compare at least three troughs or peaks, their spacing and neighbouring variations. Do not rely on one feature.

4. Read the candidate year of the first ring and inspect agreement and disagreement across all 60 rings. The curves need not coincide exactly.

5. Try another position and explain why the first candidate is preferable.

This practice is not graded. Use the existing crossdating activity slot. Introduce skeleton plotting as a method; retain earlier synthetic exercises for optional revision.

Provenance and limits

Sample A is an excerpt from a measured Koyamaki series. The teaching reference was recomputed from 26 other trees, excluding the target individual. Each series was divided by its own 1800–1899 mean; the resulting relative widths were averaged by year. This is not a direct reproduction of the published reference chronology. No age-related detrending was applied.

This is a 60-ring exercise with known dates hidden. A section containing a clear match was selected for teaching. It does not establish that visual overlay alone dates an unknown sample, or define a correlation threshold for dating. Research combines complete series, other radii and trees, anatomical checks and statistical comparisons.

Suitability for skeleton plotting cannot be determined from species names alone. Direct comparison of whole curves is the main exercise for these materials and this audience.

Sources: Yonenobu et al. (2026), Koyamaki study; research data. Physical film printing and handling have not yet been tested.